COMPUTER NETWORKS

Comprehensive Study Material

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1. OSI MODEL

Layer 1 - Physical: Deals with physical transmission of data over cable or wireless. Includes bits, voltage levels, transmission rates, and physical connections.

Layer 2 - Data Link: Responsible for reliable data transfer between adjacent nodes. MAC addressing, switching, and frame formatting occur here. Bridges and switches operate at this layer.

Layer 3 - Network: Handles logical addressing and routing. IP protocols operate here. Routers forward packets based on IP addresses. Enables communication across networks.

Layer 4 - Transport: Ensures end-to-end communication and data delivery. TCP provides reliable connection-oriented service. UDP provides unreliable connectionless service.

Layers 5-7: Session layer manages conversations, presentation layer handles encryption and compression, application layer provides user services like HTTP, FTP, SMTP.

2. TCP/IP MODEL

Comparison with OSI: TCP/IP model has 4 layers (or 5 with split) versus OSI's 7. TCP/IP is the practical model used on the Internet. Simpler but still comprehensive.

Link Layer: Corresponds to OSI layers 1-2. Handles hardware addressing, physical transmission, and media access control.

Internet Layer: Corresponds to OSI layer 3. IP, ICMP, and IGMP protocols operate here. Routing and logical addressing.

Transport Layer: Corresponds to OSI layer 4. TCP and UDP manage end-to-end communication. Port numbers identify services.

Application Layer: Corresponds to OSI layers 5-7. HTTP, HTTPS, FTP, SMTP, DNS, SSH operate here. User services and APIs.

3. IP ADDRESSING

IPv4: 32-bit addresses represented as four octets (e.g., 192.168.1.1). Supports 2^{32} addresses (~4.3 billion). Divided into classes A-E or using CIDR notation.

Subnetting: Divides networks into smaller subnets using subnet masks. Allows efficient IP address allocation. Subnet mask defines which bits identify the network.

CIDR Notation: Classless Inter-Domain Routing notation (e.g., 192.168.1.0/24). First 24 bits identify network, last 8 bits identify host. More flexible than class-based addressing.

IPv6: 128-bit addresses supporting 2^{128} addresses. Uses hexadecimal notation (e.g., 2001:0db8::1). Solves IPv4 address exhaustion and includes built-in security features.

NAT: Network Address Translation allows private IPs behind single public IP. Enables communication while conserving public addresses. Can complicate peer-to-peer connections.

4. ROUTING

Static Routing: Administrators manually configure routes. Suitable for small networks. No overhead but lacks flexibility and scalability.

Dynamic Routing: Routers exchange information and adapt paths automatically. Handles network changes and failures. Requires more processing.

RIP (Routing Information Protocol): Distance-vector protocol. Uses hop count as metric. Simple but limited to 15 hops. Obsolete but foundational.

OSPF (Open Shortest Path First): Link-state protocol. Uses Dijkstra's algorithm. Converges faster than RIP. Suitable for large networks.

BGP (Border Gateway Protocol): Used for inter-AS routing on Internet backbone. Bases decisions on policies and AS path. Complex but essential for Internet.

5. TRANSPORT LAYER

TCP (Transmission Control Protocol): Connection-oriented, reliable, ordered delivery. Establishes connection via 3-way handshake. Flow control, error detection, retransmission. Higher overhead.

UDP (User Datagram Protocol): Connectionless, unreliable, fast delivery. No connection setup. Minimal overhead. Suitable for real-time applications tolerating some loss.

Ports: 16-bit numbers identifying services. Well-known ports 0-1023, registered 1024-49151, dynamic 49152-65535. Enable multiplexing on single IP.

Flow Control: Prevents sender overwhelming receiver. Sliding window mechanism in TCP adjusts transmission rate based on receiver capacity.

6. DNS

Domain Name System: Translates human-readable domain names to IP addresses. Distributed database of domain records. Critical infrastructure of the Internet.

DNS Hierarchy: Root nameservers, top-level domain (TLD) servers, authoritative nameservers. Hierarchical structure enables scalability and fault tolerance.

DNS Records: A records map domains to IPv4, AAAA to IPv6, CNAME creates aliases, MX specifies mail servers, NS designates nameservers.

Caching: Recursive resolvers cache responses. TTL (Time To Live) controls cache expiration. Reduces load on authoritative servers and speeds up resolution.

7. HTTP/HTTPS

HTTP: Hypertext Transfer Protocol. Client-server model using TCP port 80. Stateless protocol. Methods: GET (retrieve), POST (submit), PUT (update), DELETE (remove).

Status Codes: 1xx (informational), 2xx (success), 3xx (redirection), 4xx (client error), 5xx (server error). Help diagnose request issues.

HTTPS: HTTP over SSL/TLS encryption. Uses TCP port 443. Prevents eavesdropping and tampering. Essential for sensitive data like banking and passwords.

SSL/TLS: Secure Sockets Layer/Transport Layer Security protocols. Use certificates for authentication. Establish secure channel through handshake. HTTP/2 and HTTP/3 typically use HTTPS.

8. NETWORK SECURITY

Firewalls: Filter traffic based on rules. Stateless firewalls check individual packets, stateful track connections. Block unauthorized access while allowing legitimate traffic.

Encryption: Transforms data to unreadable form. Symmetric (same key for encrypt/decrypt) and asymmetric (public/private keys) encryption. Essential for confidentiality.

Authentication: Verifies user/device identity. Passwords, multi-factor authentication, certificates. Prevents unauthorized access.

DDoS Protection: Mitigates distributed denial-of-service attacks. Rate limiting, traffic filtering, load balancing. Absorbs or redirects massive traffic floods.

9. WIRELESS NETWORKS

WiFi (802.11): Wireless LAN technology. Standards: 802.11a/b/g/n/ac/ax. Operates on 2.4GHz and 5GHz bands. WPA2/WPA3 provide security.

Cellular Networks: 4G LTE and 5G provide mobile broadband. Divided into cells served by base stations. Support mobility while maintaining connectivity.

Channel and Interference: Multiple devices use same spectrum. Channel allocation, frequency reuse, and interference management enable coexistence.

10. NETWORK TROUBLESHOOTING

Ping: Tests connectivity by sending ICMP echo requests. Measures round-trip time. Indicates if host is reachable and latency.

Traceroute: Maps path packets take through network. Shows each hop and latency. Identifies where connectivity breaks.

Netstat: Displays network statistics and connections. Shows open ports, established connections, and packet statistics.

Wireshark: Packet sniffer and analyzer. Captures and examines network traffic. Helps debug application-level issues and network problems.